

CABAMINE INJECTION 1000mcg/ml

DESCRIPTION: Cabamine Injection 1000mcg/ml is a sterile, clear, reddish solution.

COMPOSITION: Each 1 ml ampoule contains 1000 mcg Cyanocobalamine. Other ingredients include Sodium Acid Phosphate 3.12 mg, N/1HCl 0.00046 ml.

PHARMACODYNAMICS: Cyanocobalamin is the most stable and widely used form of Vitamin B12, and has haematopoietic activity apparently identical to that of the anti-anemia factor in purified liver extract. Parenteral (intramuscular) administration of Vitamin B12 completely reverses the megaloblastic anemia and GI symptoms of Vitamin B12 deficiency. Gastrointestinal absorption of Vitamin B12 depends on the presence of sufficient intrinsic factor and calcium ions. Intrinsic factors deficiency causes pernicious anemia, which may be associated with subacute combined degeneration of the spinal cord.

In addition, cyanocobalamine and hydroxocobalamine are used in the Schilling test to investigate vitamin B12 absorption deficiency states. They are given in a non-radioactive form together with cyanocobalamin radioactively-labelled with cobalt- 57 (Cobalt-57) or cobalt-58 (Cobalt-58) and the amount of radioactivity excreted in the urine can be used to assess absorption status. A differential Schilling test, in which the forms of cyanocobalamine are given under different conditions (such as with intrinsic factor, antibacterials, or pancreatic enzymes) can provide information about the cause of the malabsorption. A single intramuscular injection is given to temporarily saturate B12 receptors in the liver with enough normal vitamin B12 to prevent radioactive vitamin B12 binding in body tissues (especially in the liver), so that if absorbed from the G.I tract, most of the radioactive vitamin B12 will pass into the urine. The patient's urine is then collected over the next 24 hours to assess the absorption. A normal result shows at least 5% (8-40%) of the radiolabeled vitamin B12 in the urine over the first 24 hours. In patients with pernicious anemia or with deficiency due to impaired absorption, less than 5% of the radiolabeled vitamin B12 is detected.

PHARMACOKINETICS:

Absorption

Bioavailability:

- Intramuscular, injection: rapid and complete.
- Following 100 mcg and 1000 mcg parenteral doses of cyanocobalamine, total body retention is reportedly 55% and 15%, respectively. The remainder of these doses is excreted unchanged in the urine.

Distribution

- Vitamin B12 binds in plasma to transcobalamin II, a beta-globulin, and this complex is transported to tissues (liver, bone marrow, endocrine glands, kidney).
- LIVER: 90%
- Preferential distribution to hepatic parenchymal cells is observed, and the liver serves as a storage site for other tissues. Up to 90% of body stores are in the liver (1 to 10 mg), where the vitamin is stored as the active coenzymes with a turnover rate of 0.5 to 8 micrograms daily.

Metabolism

Vitamin B12 is metabolized in the liver and has a half-life of about 6 days.

Excretion

- Renal Excretion 50% to 98%
- Between 50% and 98% of an intramuscular dose (100 to 1000mcg) of cyanocobalamine is excreted unchanged in the urine, most within the first 8 hours post injection.
- Between 1 and 3 mcg of vitamin B12 is excreted daily in the bile. Up to 50% of this amount is reabsorbed, establishing an enterohepatic recirculation of the vitamin. In patients with pernicious anemia (or following total gastrectomy), enterohepatic recirculation is impaired due to lack of intrinsic factor which results in continuous depletion of hepatic stores of vitamin B12.

INDICATION:

Cabamine Injection 1000mcg/ml is used:

- For prophylaxis of anaemia associated with Vitamin B12 deficiency.
- For uncomplicated pernicious anaemia or Vitamin B12 malabsorption.
- For Schilling Test as a diagnostic aid to investigate Vitamin B12 malabsorption.

RECOMMENDED DOSAGE:

i) Prophylaxis of anaemia:

250-1000 mcg intramuscularly every month.

ii) Uncomplicated pernicious anaemia or Vitamin B12 malabsorption:

Initial 100 mcg daily for 5-10 days followed by 100-200 mcg monthly until complete remission is achieved.

Maintenance: up to 1000 mcg monthly.

CHILD 30-50 mcg daily for 2 or more weeks (to a total dose of 1-5mg) OR AS PRESCRIBED.

iii) Schilling Test:

The flushing dose for Schilling test is 1000 mcg (1mg) given intramuscularly.

ROUTE OF ADMINISTRATION: Intramuscular.

CONTRAINDICATIONS: Hypersensitivity to cobalt and B12 products.

WARNINGS & PRECAUTIONS: Cabamine Injection 1000mcg/ml should not be administered intravenously, although small amounts of cyanocobalamine are sometimes included in total parenteral nutrition (TPN) solutions. There can be false positive results for Schilling test. The most common reason for this is inadequate urine collection. Other reasons include renal failure and dysfunction of the lining of the small intestine secondary to B12 deficiency. Cyanocobalamine or hydroxocobalamine should, if possible, not be given to patients with suspected vitamin B12 deficiency without first confirming the diagnosis. Regular monitoring of the blood is advisable. Cyanocobalamine should not be used for Leber's disease or tobacco amblyopia since these optic neuropathies may degenerate further. A sensitivity history should be obtained from the patient prior to administration of Vitamin B12; an intradermal test dose is recommended before Vitamin B12 is administered for the management of vitamin deficiency to patients who may be sensitive to cobalamins.

Serum potassium concentrations should be monitored during early Vitamin B12 therapy and potassium administered if necessary, since fatal hypokalemia could occur upon conversion of megaloblastic anemia to normal erythropoiesis with Vitamin B12 as a result of increased erythrocyte potassium requirements. Because

Vitamin B12 deficiency may suppress the signs of polycythemia vera, treatment with cyanocobalamin may unmask this condition. The increase in nucleic acid degradation produced by administering Vitamin B12 to Vitamin B12-deficient patients could result in a goal in susceptible individuals. Therapeutic response to Vitamin B12 may be impaired by concurrent, or by drugs having bone marrow suppressant effects (e.g. chloramphenicol).

Folic acid should be administered with extreme caution to patients with undiagnosed anaemia by alleviating hematologic manifestations to progress. This may result in severe nervous system damage before the correct diagnosis is made. Vitamin preparations containing folic acid should be avoided by patients with pernicious anaemia because folic acid may actually potentiate neurologic complications of Vitamin B12 deficiency. Doses of cyanocobalamin exceeding 10 mcg daily may improve folate-deficient megaloblastic anaemia and obscure the true diagnosis. Folate may mask vitamin B12 deficiency.

INTERACTION WITH OTHER MEDICAMENTS: Serum concentration of cyanocobalamin may be lowered by oral contraceptives but this interaction is unlikely to be significant but should be taken into account when performing assays for blood concentrations. Parenteral chloramphenicol may attenuate the effect of Vitamin B12 in anaemia.

Meanwhile, previously administered diagnostic and therapeutic radio pharmaceuticals may interfere with performance of the Schilling test and interfere with interpretation of the test for prolonged periods of time.

USE IN PREGNANCY & LACTATION:

Pregnancy: Vitamin B12 requirements are increased in pregnant women. Parenteral preparations should be used during pregnancy only when the potential benefits justify the potential risks to the fetus.

Lactation: Vitamin B12 is distributed into human milk, therefore, it is not recommended for breastfeeding mothers unless the expected benefits to the mother outweigh the potential risk to the infant.

SIDE EFFECTS: Allergic hypersensitivity reactions have occurred rarely after cyanocobalamin and include skin rashes such as rash and itching, and anaphylaxis. Patients who are hypersensitive to cyanocobalamin injection may be able to take oral cyanocobalamin. Injection site reactions including pain, erythema, pruritus, induration, swelling, and necrosis can occur.

Other adverse effect reported with cyanocobalamin include gastrointestinal disturbances, fever, chills, hot flushing, dizziness, malaise, acneiform and bullous eruption, and tremor.

SYMPTOMS AND TREATMENT OF OVERDOSE: Overdosage with Cyanocobalamin is quite unlikely.

INCOMPATIBILITIES: Vitamin B12 injection has been reported to be incompatible with chlorpromazine hydrochloride, phytonadione, prochlorperazine, edisylate, warfarin sodium, ascorbic acid, dextrose, heavy metals, oxidizing or reducing agents, and alkaline or strongly acidic solutions.

STORAGE CONDITIONS: Store below 30°C. Protect from light: Store ampoules in the original outer carton until the ampoules have been used. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE: Please refer to outer package.

PACK SIZE: Pack in box of 10's in 1 ml ampoules

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59 Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

15000